



University
of Exeter

COMM117

Large Language Models and Application

Previously

An LLM is just a probabilistic next-token predictor

- Input → text
- Model → probabilities
- Output → sample next token
- Repeat

Previously

- LLM ideas are not new
- The breakthrough is **scale**
- Three things **scaled together**:
 - Data
 - Compute
 - Architecture

Nothing here is magic — it's **scale** + the right **inductive bias**.

Previously

- LLM ideas are not new
- The breakthrough is **scale**
- Three things **scaled together**:

➤ Data

➤ Compute

➤ Architecture

Nothing here is magic — it's **scale** + the right **inductive bias**.

Introduction Today

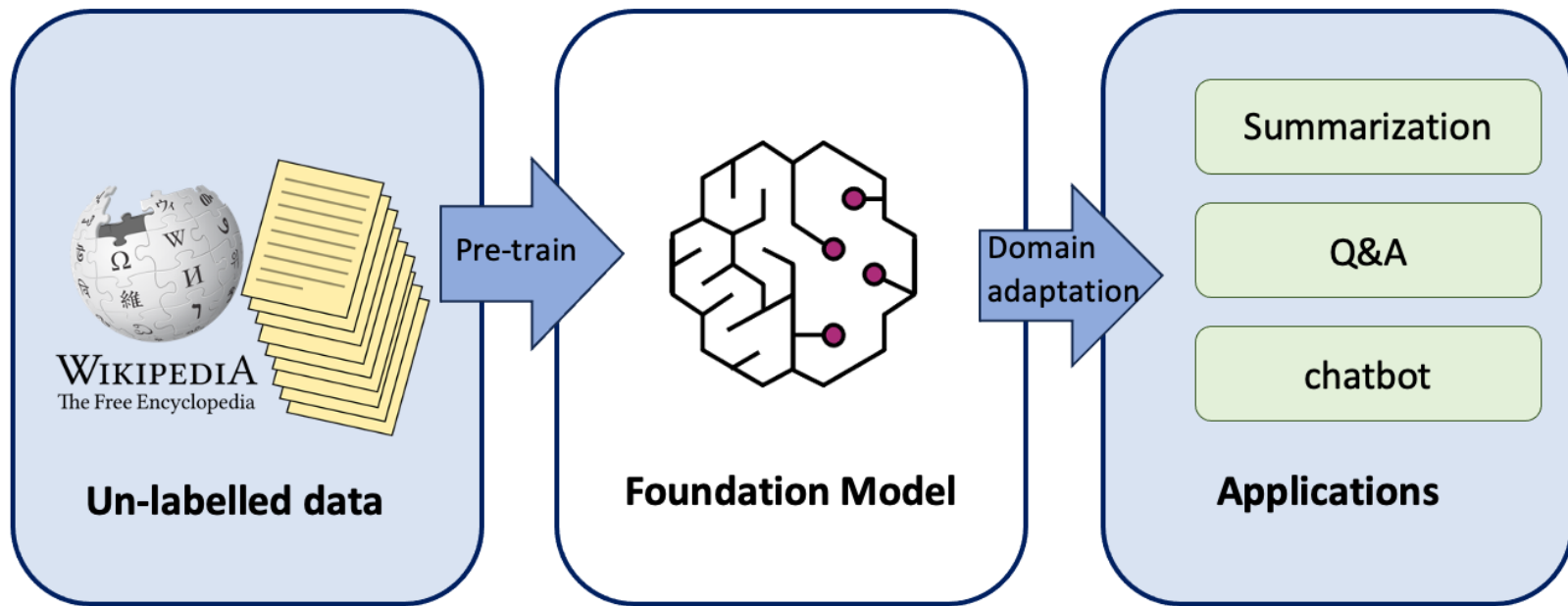
- Data Collection
- Data pre-processing
 - Data cleaning
 - Filtering
- Data chosen impact

Data

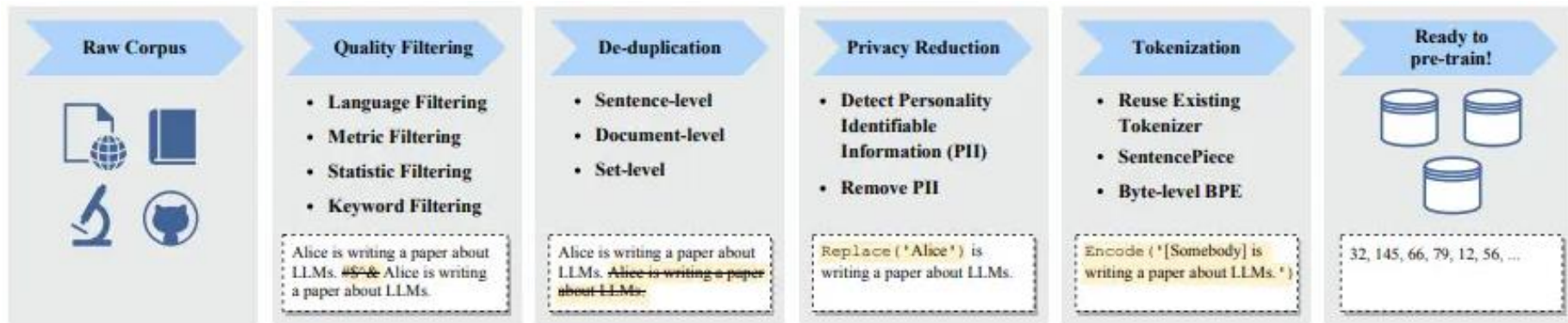
Data: learning from the internet

- **Web-scale** text (books, articles, code, forums)
- Orders of magnitude **more data** than earlier NLP
- **Language** as a self-supervised signal ----
---- The data labels come from the data itself.

Training Pipeline for LLM



Data Preprocessing Pipeline



Raw data collection

LLMs require **higher-quality** and **larger-scale** data than smaller language models

LLM performance depends heavily on:

- The **source and diversity** of pretraining data
- The scale and **quality** of the corpus
- The **preprocessing and filtering strategies** applied

Raw data collection

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- The **source and diversity** of pretraining data
- The scale and **quality** of the corpus
- The **preprocessing and filtering strategies** applied

Compared to smaller models, LLMs are: **More sensitive to noise and duplication**

Raw data collection

Why High-Quality Public Web Data Matters

Extensive knowledge

- Exposure to a **wide range of topics** and domains
- More **accurate** and contextually relevant **responses**

Diverse writing styles and perspectives

- Coverage of **different formats** (articles, conversations, documentation)

Up-to-date information

- Public web content is **frequently updated**
- Models remain aligned with **current knowledge and trends**

Raw data collection

Why High-Quality Public Web Data Matters

Extensive knowledge

- Exposure to a **wide range of topics** and domains
- More **accurate** and contextually relevant **responses**

Data diversity and quality are critical for reducing hallucinations and improving factual grounding

Diverse writing styles and perspectives

- Coverage of **different formats** (articles, conversations, documentation)

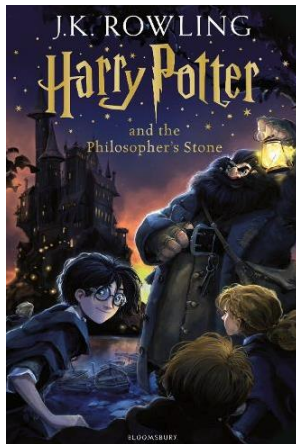
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Data Source

Raw Data

General Data



Web pages
(blog posts, reviews, Google, Amazon...)

Books

Community networks – conversation
(X, facebook, ins...)

Wikipedia

Google

data collection for LLM

AI Mode All Images Videos Shopping Short videos News More Tools

Turing
<https://www.turing.com/resources/data-collection-met...>

Prominent Data Collection Methods and Tools for LLMs

2 Apr 2025 — In this blog, we explore different data collection sources and tools, and help you navigate various challenges associated with data sourcing for LLM training. [Read more](#)

Wikipedia
<https://www.wikipedia.org>

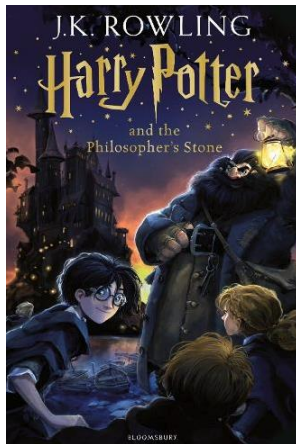
Wikipedia

Wikipedia is a free online encyclopedia, created and edited by volunteers around the world and hosted by the Wikimedia Foundation.

Data Source

Raw Data

General Data



Web pages
(blog posts, reviews, Google, Amazon...)

Books

Community networks – conversation
(X, facebook, ins...)

Wikipedia

Abundant, diverse, and easily accessible



Wikipedia

<https://www.wikipedia.org>

Wikipedia

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data collection for LLM



AI Mode

All

Images

Videos

Shopping

Short videos

News

More

Tools



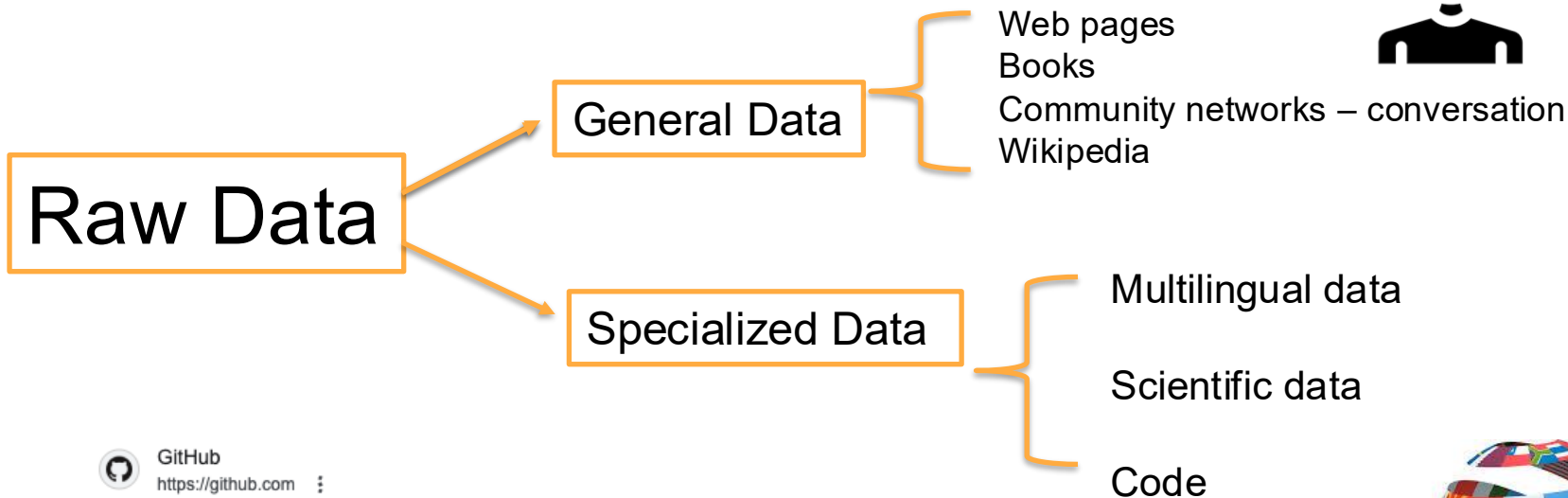
Turing

<https://www.turing.com/resources/data-collection-met...>

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Data Source



 GitHub
<https://github.com>

GitHub · Change is constant. GitHub keeps you ahead. · GitHub

Give LLMs specific problem-solving abilities



Reality Check: Raw Web Data Is Messy

Raw web text contains:

- HTML tags
- Duplicates
- Spam & SEO text
- Hate speech & toxicity
- Misinformation
- Private or sensitive data



Data Preprocessing Is Necessary

Raw web text contains:

- HTML tags
- Duplicates
- Spam & SEO text
- Hate speech & toxicity
- Misinformation
- Private or sensitive data



Without preprocessing:

- Models learn **noise**
- **Bias** is amplified
- **Safety** risks increase
- Training becomes **inefficient**

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AI (artificial intelligence)

This article is more than 1 month old

Artificial intelligence research has a slop problem, academics say: 'It's a mess'

AI research in question as author claims to have written over 100 papers on AI that one expert calls a 'disaster'



The author, Kevin Zhu, now runs Algoverse, an AI research and mentoring company for high schoolers. Photograph: Cavan Images/Alamy

A single person claims to have authored 113 academic papers on artificial intelligence this year, 89 of which will be presented this week at one of the world's leading conferences on AI and machine learning, which has raised

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Aisha Down

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Data from Web

```
<!doctype html>
<html lang="en">
<head>
  <title>AI Breakthrough Shocks Researchers</title>
  <meta charset="utf-8">
  <script src="/ads/track.js"></script>
  <style>
    .hidden { display:none; }
  </style>
</head>

<body>
  <div id="cookie-banner">
    We use cookies to personalise content and ads.
    <button>Accept all</button>
    <button>Manage settings</button>
  </div>

  <header>
    <nav>
      Home | World | Tech | Politics | Sports | Subscribe | Sign In
    </nav>
  </header>
```

Without preprocessing

AI Breakthrough Shocks Researchers

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Scientists announced a new AI model that achieves
state-of-the-art results

Contact the editor momo@exiii.ac.com

After Preprocessing



AI Breakthrough Shocks Researchers

Scientists announced a new AI model that achieves
state-of-the-art results
in language tasks.

What is data preprocessing

Data preprocessing = transforming raw text into **clean, usable** training data

Includes:

- Cleaning
- Normalisation
- Filtering
- Deduplication
- Privacy reduction
- Quality control

Step 1: Cleaning Text

Typical cleaning steps:

- Remove Noise:
HTML / boilerplate, Eliminate unwanted symbols, emojis, hashtags, and Unicodes
- Strip scripts & navigation text
- Normalize whitespace and Convert the text to lowercase for consistency.
- Remove broken encoding

Step 1: Cleaning Text

```
def basic_clean(text):  
    text = re.sub(r"[\s*\d+\s*\]", "", text)      # remove [1], [ 12 ]  
    text = re.sub(r"\n{3,}", "\n\n", text)      #Collapse excessive newlines  
    text = re.sub(r"[\t]{2,}", " ", text)        #Normalize multiple spaces and tabs  
    lines = [ln.strip() for ln in text.splitlines()] #Split into lines & strip whitespace  
    lines = [ln for ln in lines if ln]           #Drop empty lines  
    return "\n".join(lines)
```

Natural language processing is a field. [1]

It focuses on language understanding.

Applications include translation.



Natural language processing is a field.

It focuses on language understanding.

Applications include translation.

Step 2: Normalise

Convert messy, inconsistent text into a **single, clean, space-normalized string** suitable for tokenisation and modelling.

```
WHITESPACE_RE = re.compile(r"\s+")
NONPRINT_RE = re.compile(r"[\x00-\x08\x0B\x0C\x0E-\x1F]")

def normalize(text):
    x = NONPRINT_RE.sub(" ", text)
    x = x.replace("\u00a0", " ") # non-breaking space
    x = WHITESPACE_RE.sub(" ", x).strip()
    return x
```

ASCII
non-printable control characters

"Natural\u00a0language\u00a0processing"



"Natural language processing"

Step 3: Filtering

Why language filtering?

- LLMs are often language-specific
- Mixed-language data hurts performance

Common approaches:

- Language ID models
- Character distribution heuristics

Step 3: Filtering

Length filtering

Remove texts that are:

- Too short (no signal)
- Too long (logs, dumps)

Why?

Short text → low information

Extremely long text → noise & compute waste

Step 3: Filtering

```
def length_filter(text, min_chars, max_chars) :  
    return (min_chars <= len(text) <= max_chars)
```

```
def heuristic_quality_filter(text):  
    boilerplate_patterns = [  
        r"cookies?",  
        r"privacy",  
        r"terms",  
        r"sign in",  
        r"log in",  
        r"copyright",  
        r"download as pdf",  
        r"printable version",  
        r"edit",  
        r"talk page",  
        r"citation",  
        r"verification",  
    ]
```

```
    bp = re.compile("|".join(boilerplate_patterns), re.IGNORECASE)  
  
    kept = []  
    for ln in text.splitlines():  
        if len(ln) < 40:  
            continue  
        if bp.search(ln):  
            continue  
        kept.append(ln)  
    return "\n".join(kept)
```

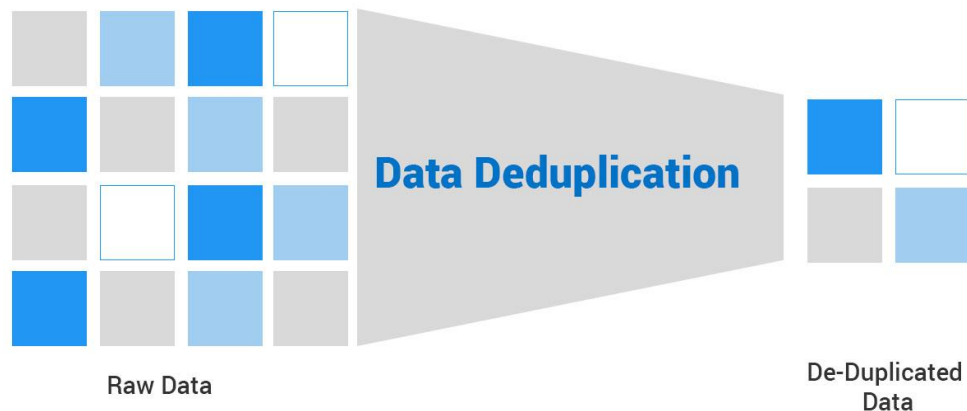
Step 4: Deduplication

Why duplicates are bad:

- Overfitting
- Memorisation
- Biased frequency statistics

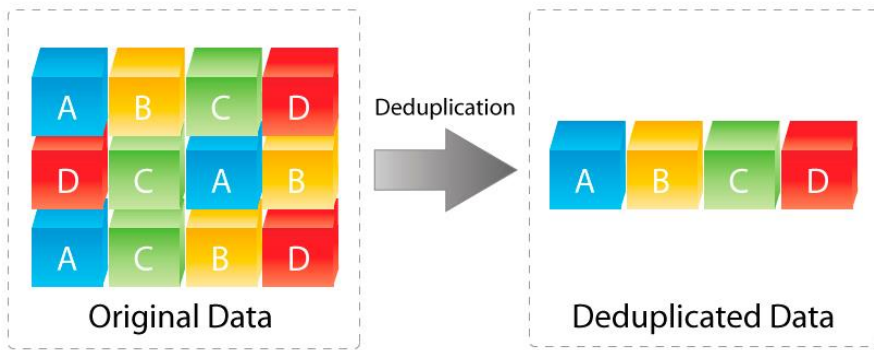
Sources of duplication:

- Copy-paste websites
- Mirrors
- Quoted content



Step 4: Deduplication

```
def stable_hash(text) :  
    return hashlib.sha256(text.encode("utf-8")).hexdigest()  
  
def exact_dedup(texts):  
    """  
    Returns unique texts + mapping from original index -> unique index  
    """  
    seen = {}  
    unique = []  
    mapping = {}  
    for i, t in enumerate(texts):  
        h = stable_hash(t)  
        if h not in seen:  
            seen[h] = len(unique)  
            unique.append(t)  
        mapping[i] = seen[h]  
    return unique, mapping
```



Step 5: Privacy reduction

```
EMAIL_RE = re.compile(r"\b[\w\.-]+@[ \w\.-]+\.\w+\b")
PHONE_RE = re.compile(r"\b(\+?\d[\d\s-]{7,}\d)\b")

def redact_pii(text: str) -> str:
    x = EMAIL_RE.sub("[REDACTED_EMAIL]", text)
    x = PHONE_RE.sub("[REDACTED_PHONE]", x)
    return x
```

Data quality

More data is always better?

Data quality

More data is always better? **No!**

High-quality + diverse data





Small clean dataset vs massive noisy dataset



Small Clean vs Massive Noisy

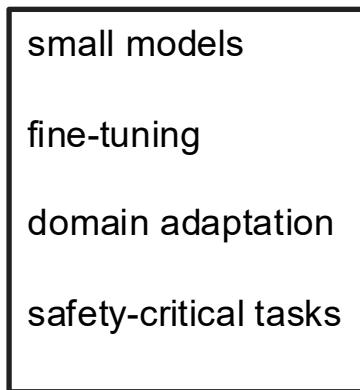
Small, Clean Dataset

Pros

- High signal-to-noise ratio
- Faster training

Cons

- Limited coverage
- Poor generalisation
- Overfitting risk
- Knowledge gaps



Small Clean vs Massive Noisy

Small, Clean Dataset

Pros

- High signal-to-noise ratio
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- Knowledge gaps

Best for:

Fine-tuning

Specialized domains (medical, legal)

Evaluation and alignment

Low-resource settings

Small Clean vs Massive Noisy

Massive, Noisy Dataset

Pros

- broad coverage of language and knowledge
- Enables emergent abilities
- Better generalisation
- Supports scaling laws

Cons

- Noise slows learning
- Bias and toxicity leakage
- Higher compute cost
- Memorisation risk

Small Clean vs Massive Noisy

Massive, Noisy Dataset

Pros

- broad coverage of language and knowledge
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Best for

Pretraining large LLMs

General-purpose language understanding

Few-shot and zero-shot capabilities

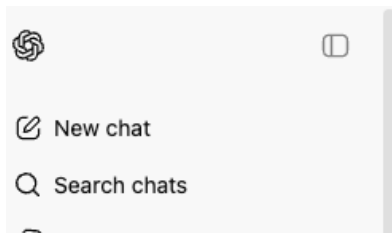
Small Clean vs Massive Noisy

Massive, lightly filtered data $\longrightarrow$ learn language broadly
Smaller, high-quality data $\longrightarrow$ refine behaviour

This is exactly what:

- GPT-style models
- LLaMA-style models
- PaLM-style models

do in practice.



Dataset Impact--- something bad

- Bias
- Harmful content
- Privacy leakage
- ...

Bias in Training Data

“A **nurse** walked into the room. **She** said...”

“An **engineer** walked into the room. **He** said...”

“Ronghui Mu is a **lecturer** in University of Exeter. **His** research focused



Models reflect the data they see — not the world as it should be.

Bias in Training Data

Where bias comes from:

- Cultural
- Geography
- Language dominance
- Historical inequalities

Bias is not inserted by the model — it is *learned from data frequency*

Toxicity & Harmful Content

Examples of harmful data:

- Hate speech
- Harassment
- Extremism
- Disinformation

“How can I lose weight quickly?”

--- Encourages extreme dieting

---- Unsafe medical suggestions

Filtering strategies:

- Blocklists
- Classifiers
- Human review (expensive!)

Filtering is imperfect but necessary

Privacy and data Leakage

Risks:

- Personal data in training sets
- Memorisation of rare sequences

Consequences:

- Legal risk
- Ethical risk
- Trust loss

“What is John Smith’s age?”



This comes back in Responsible AI (Week 10).

What happens if we train on:
Reddit only?
Academic papers only?
Code only?



What happens if we train on:

Reddit only?

Academic papers only?

Code only?



Reddit-only training

- **Fluent**, conversational language
- Strong opinions and cultural **bias**
- **Informal tone** and slang
- **Weak factual** reliability

Academic-only training

- **Precise** and cautious language
- **Strong technical** explanations
- **Poor** everyday **conversation**
- **Narrow** domain knowledge

Code-only training

- Excellent syntax and **logic**
- **Rigid** and literal reasoning
- Weak natural language **understanding**
- Poor performance on **open-ended tasks**

How Data Choices Affect LLM Behaviour

Data affects:

- Fluency
- Factuality
- Bias
- Toxicity
- Generalisation

Tokenization

“Tokenization is the process of **creating a digital representation** of a real thing.

Tokenization can also be used to protect sensitive data or to efficiently process large amounts of data.”



Next steps

- Tokenisation and NLP pipeline

Additional material:

(Paper) Vaswani et al. (2017) — Attention Is All You Need

Jurafsky & Martin — Speech and Language Processing (Draft Chapters on Neural Models)

Sebastian Raschka — LLMs from Scratch (<https://github.com/rasbt/LLMs-from-scratch>)

More on ELE!

▼ Reading List

Reading List

▼ Coursework 1

coursework specification

CW1_dataset

Coursework 1- updated

▼ Coursework 2

Coursework 2 description

Coursework 2 -updated

▼ Introduction to Python

Introduction to Python 1&2

Introduction to Python 3&4

How_to_use_JupyterNotebook

▼ Week 1 - 12 Feb

Lecture 1.1 - Introduction to L...

Lecture 1.2- Data preprocessi...

▼ Section 6

▼ Section 7

- To format citations or references

When writing your assessment, you must never use AI tools:

1. For uses other than those represented by checked boxes in the list above.
2. To translate more than a word or short phrase into English unless agreed above.
3. To upload sensitive or identifying material to an AI tool.
4. To present material that has been generated by AI as your own work or the work of someone else..

When submitting your assessment, you must:

1. Check the points during the submission process, that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.
2. Treat the AI tool like a citation from any other source.
3. Include a list of all AI prompts and hyperlinks to their output with your references, at the end of your work. You do not need to include the outputs themselves, just the links.
4. Retain the full outputs generated by the prompts you have used during your assignment. These outputs should be accessible at the hyperlinks which you have submitted with your assignment. You may be asked to produce this material in the event of an academic conduct inquiry.



[coursework specification](#) PDF



[CW1_dataset](#) TXT



[Coursework 1- updated](#) IPYNB

Thank you!

If you want to ask us any questions about the module, the university, or anything else, drop us an email.

Pro tip: write **COMM117** on the email subject line, to make sure we give it priority.

 r.mu2@exeter.ac.uk

Questions?

